



THE FASTEST WAY FOR TEAMS
TO BUILD ML VISION

DATATURE PLATFORM OVERVIEW

Build Robust Computer Vision

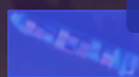
PRESENTED TO

Datature Community

Build Robust Computer Vision

Datature is an MLOps platform that helps companies and researchers build highly robust computer vision models using a mixture of AI-Assisted Labelling, Synthetic Image Generation, Automated Cloud Orchestration, and APIs

BUILD YOUR NEXT
DISRUPTIVE AI WITH
DATATURE →

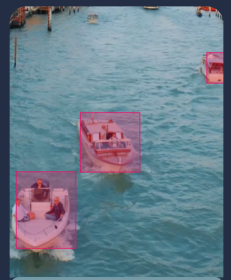


SHIP

UNKNOWN BOAT



ANOMALOUS
STRUCTURE



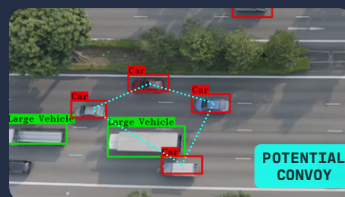
MULTIPLE SHIPS
DETECTED AT PORT C



DEFORESTATION
ACTIVITIES



32500 SQFT OF ROOF



POTENTIAL
CONVOY



DEFECTIVE



DEFECTS DETECTED IN
CROSS SECTION

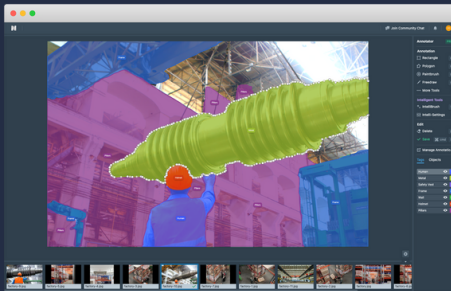


POSSIBLE
ANOMALY

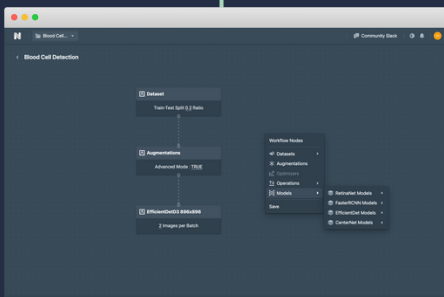


CONSTRUCTION SITE FULL

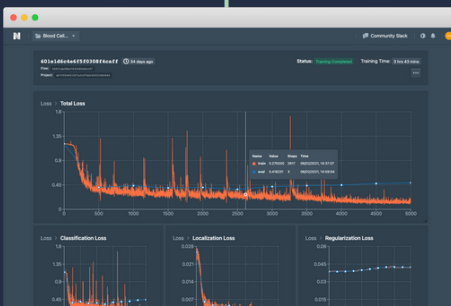
PLATFORM FLOW



Collaboratively label complex data with Intellibrush and leverage Nexus as your ground-truth / asset manager



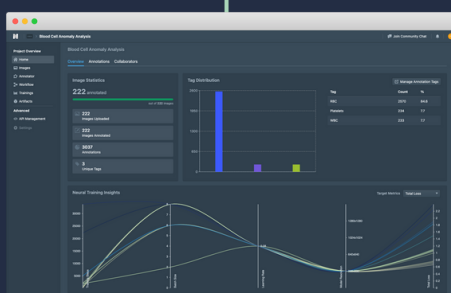
Leverage augmentations, preprocessing, over 30+ neural networks, and multi-GPU training just by connecting nodes



Monitor for overfitting, non-converge, and evaluation to determine performance of models in real-time



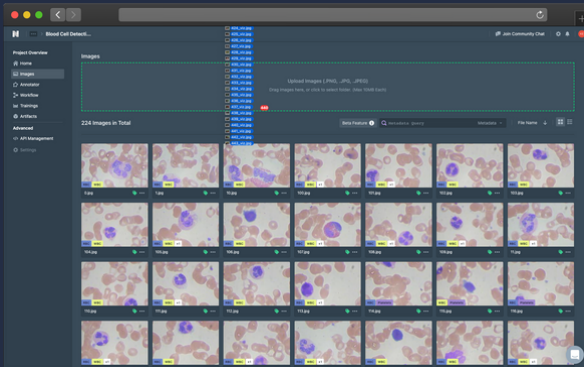
Export Tensorflow models, Host model on Datature API, or deploy on your own cloud -- your model, your choice



Add new data, run new transfer learning sessions, and monitor for long-term improvements. Iterate.

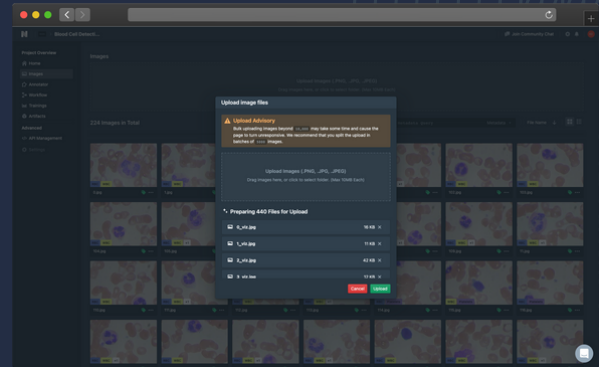
An Entire MLOps Toolkit Within Your Browser

ONBOARDING DATA



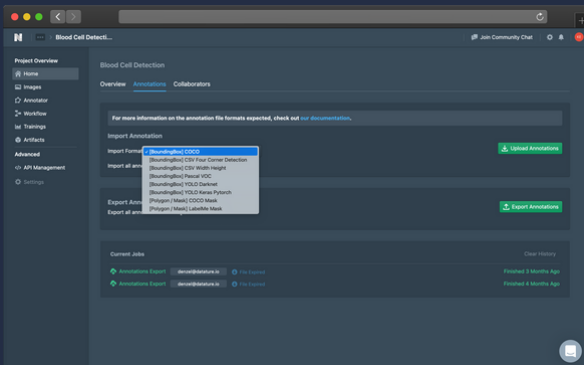
Adding Data

After creating a new project, users can navigate to *Images* tab and drag images into the upload zone. You can drag an entire folder and the system will filter the images for uploading.



Uploading Images

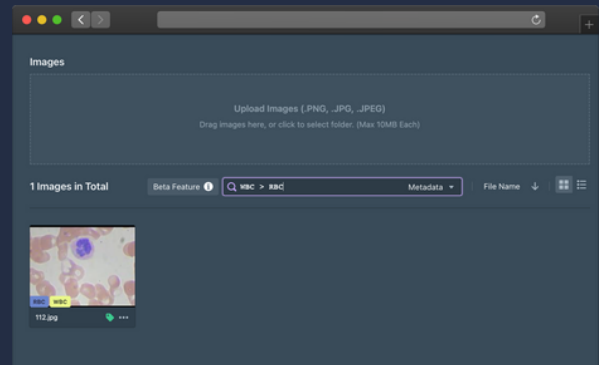
Datature uses SignedURLs to securely upload your assets. We support PNG, JPGs under 10MB in size each for now.



Adding Annotations

If you have existing annotations from other tools - you can upload them by selecting the matching format.

Note that the images have to be uploaded first for the system to match your files with the assets.



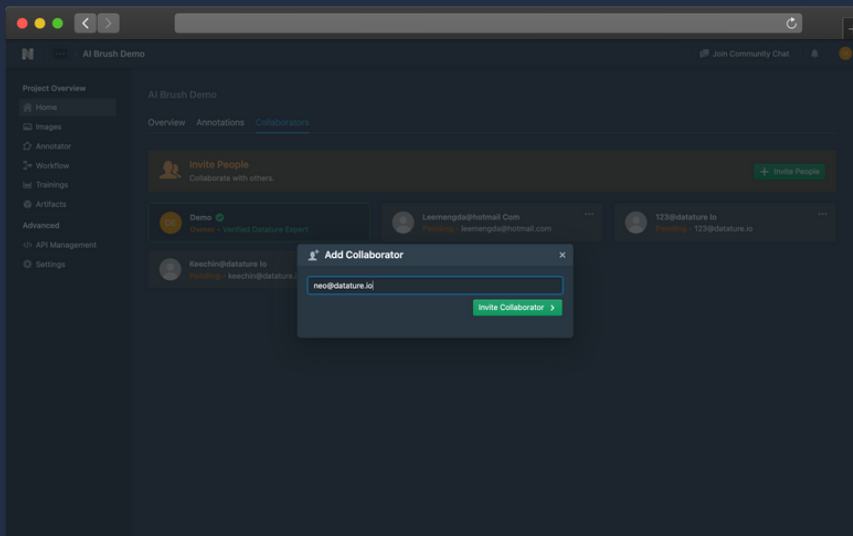
Searching Assets

You can use *Metadata Query* to check the sanity of your data using expressions such as

`WhiteCells > RedCells`

This will return images where there are more white cells than red - which might be a sign of mis-labelling.

ONBOARDING TEAM

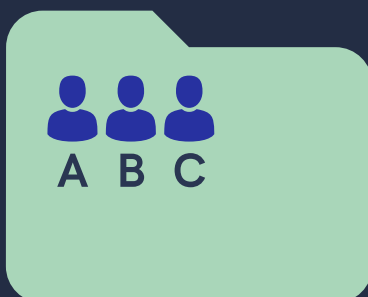


Adding Team Members

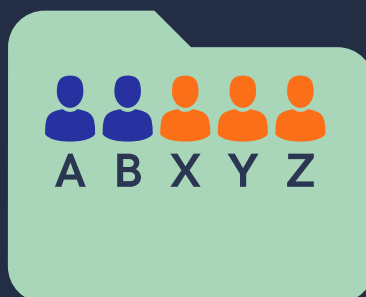
Depending on your subscribed plan, you are able to invite your team to collaborate on various tasks such as data labelling, monitoring training, exporting files.

You can navigate to the *Collaborator* tab under *Home* of your project. Each project have separate team structure.

Team members need to sign up for at least a Free Plan before they can be invited.



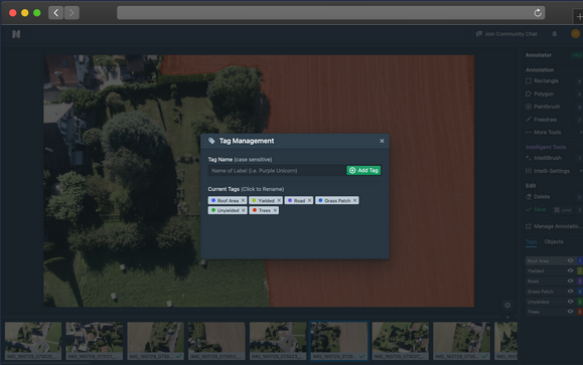
Project A



Project B

Total Collaborator
Quota Used → 6

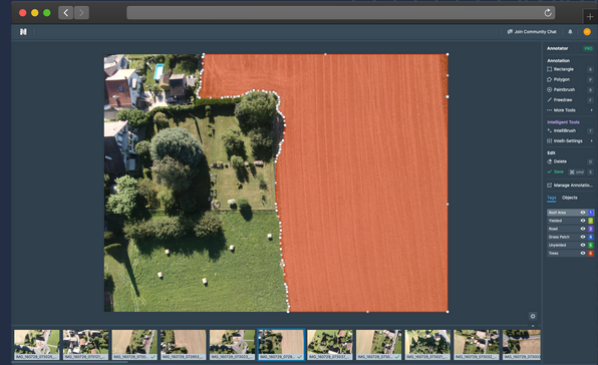
ANNOTATING DATA



Creating Tags

Tags is a fundamental label, such as *Dog*, *Cat*, etc. To create a tag, select *Manage Annotation* button on the right panel.

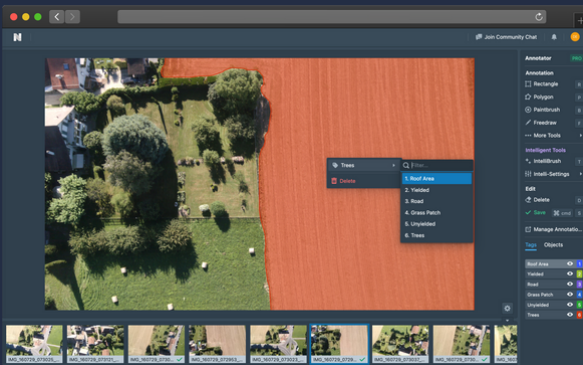
Renaming a tag to an existing tag will merge the tags together.



Drawing Annotations

Datature feature a myriad of tools to draw both bounding boxes and masks.

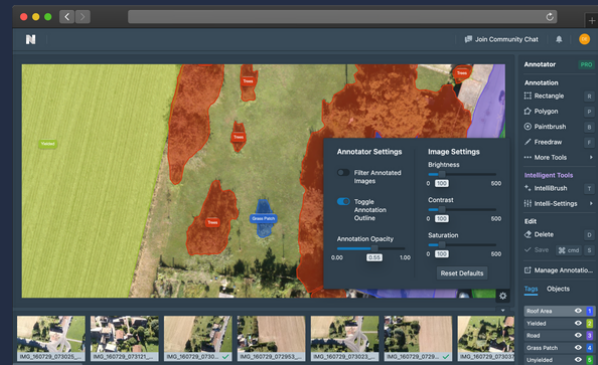
Simply select a tag, or use the numerical hotkey shortcut, before drawing your shape.



Editing Annotations

To make minor edits to a shape, click on the shape and you will be able to move the vertices.

To edit other properties, simply right-click the shape and an edit menu will appear.



Annotator Settings

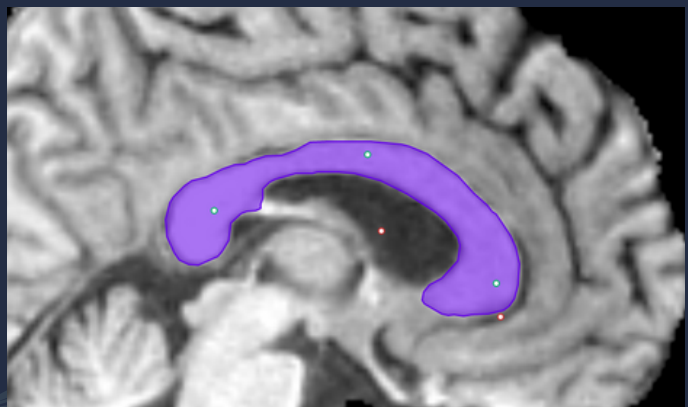
Press the *Cog* (Setting) button to change the settings such as brightness, filtering images annotated by the team, etc.

This does not affect the image in anyway and is purely aesthetics.

INTELLIBRUSH

Intellibrush uses state-of-the-art deep learning and computer vision algorithms to allow users to quickly label highly complex data in just a few clicks.

All without model pretraining.



Label complex shape by selecting **IN** and **OUT** regions

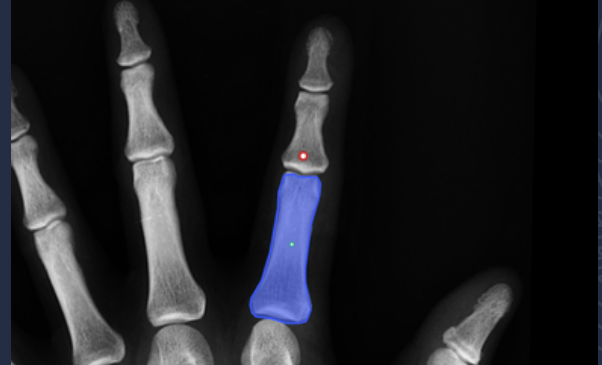
USING INTELLIBRUSH



Initiating Intellibrush

Start by activating Intellibrush, using "T" on the keyboard. The brush defaults to *IN* mode.

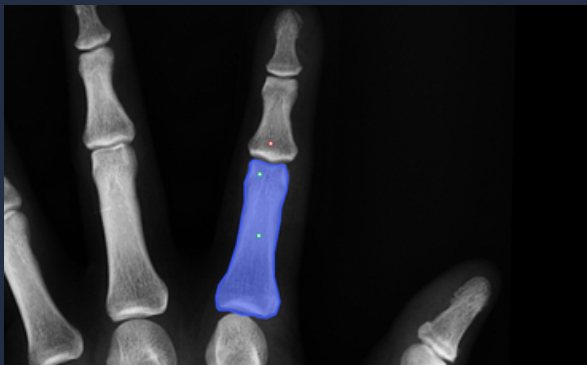
Select a region you are interested in by left-clicking on the area. In this case, we want the phalanx.



Rejecting Area

Since we only want the bottom portion, simply right-click the top region to reject the area.

Right clicking sets an *OUT* point. You can remove any points by clicking on it.



Refining Shape

We notice in the previous image that the bounds are not tight enough. To fix this, simply click another point in the region you'd like closer to the boundary as shown.



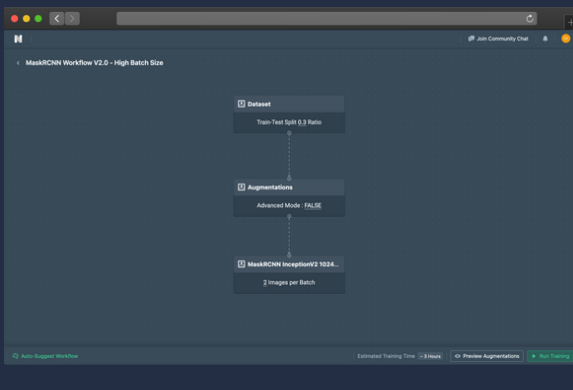
Confirming Selection

Once you are done, you should press [SPACEBAR] to commit a shape. This will fix your shape in place and consume 1 x Intellibrush Quota.

MODEL TRAINING OVERVIEW

Data Preprocessing, Synthetic Image Augmentation, Multi-GPU Strategies

Raw Images + Labels + Neural Network → Image Recognition



Create multiple experiments leveraging the RCNN family of models, EfficientDet, YOLO, etc to find the best trade-off between speed and accuracy.

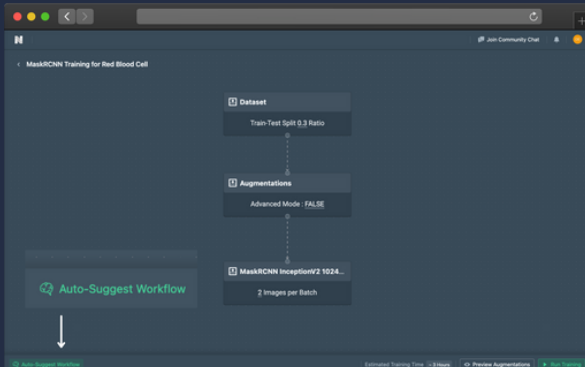
By connecting simple nodes, you can create powerful augmentations, checkpoint strategies, and leverage up to 16x Tesla V100 GPU to train your models with large batch sizes.



Monitor losses in real-time or export metrics from model training for research / trial clearance - Datature provides all the information surrounding your model training performance.

Datature will be launching *Advanced Analytics* in Q3 '22 to provide logs, visualizations on gradient activations, and where the model is looking at before arriving at a prediction.

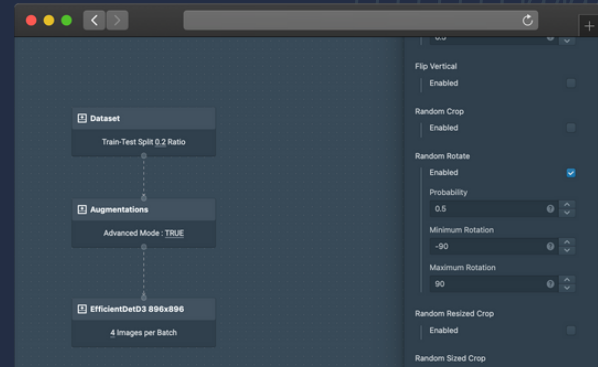
TRAINING A MODEL



Creating a Pipeline

By going under *Workflow > Create Workflow*, you can design a training session easily.

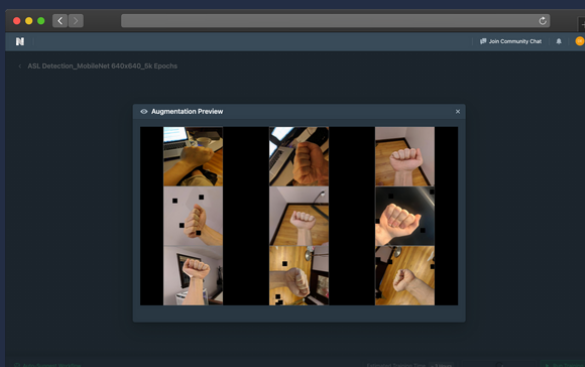
For automated suggestions, click "Auto-Suggest Workflow" at the bottom right of the screen.



Designing Augmentations

Augmentations create synthetic variations of your data to reduce odds of overfitting.

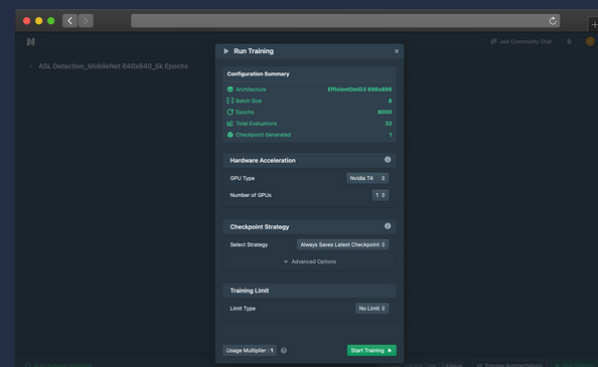
A car image, when flipped horizontally, is still a car. Variations like these helps the model learn concepts and visuals robustly.



Visualize Augmentations

Verifying your augmentation strategy's sanity is important. Various objects, when flipped around or changes color ceases to be the same class.

Use the Augmentation Preview button at the bottom right to visualize this.



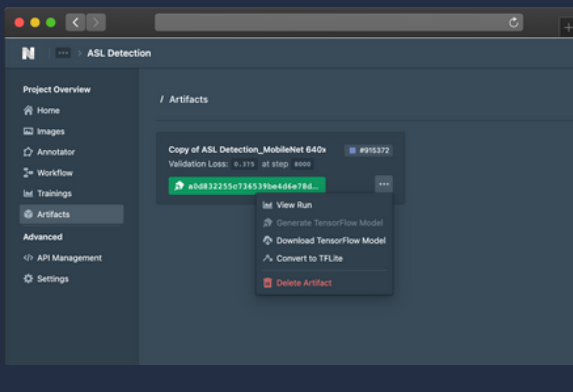
Initialize Training

Click Start Training to bring up a panel where you can edit hyperparameters.

When a training is started - a secure, sandboxed instance will be created and the neural network will begin its learning process with your hyperparameters.

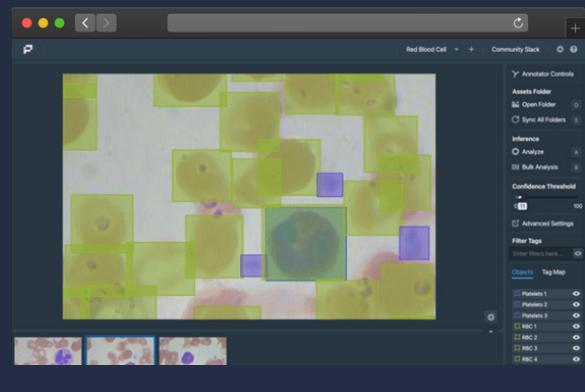
MODEL DEPLOYMENT

Analyzing Models, Exporting, and Deploying to the Edge



Datature platform allows you to export your own model in various formats to deploy on your own cloud or edge devices. We will be adding more formats such as PyTorch, ONNX, etc to support wider deployment use cases later this year.

Datature will be **launching Neural APIs in Q3 '22** to allow users to deploy API using our servers where one can simply send an image to our endpoint and received predictions and labels in less than a second.



Portal is an open-source model loader and visualizer available on [GitHub](#) where the team can easily load an exported model and some test images to see how to model is performing.

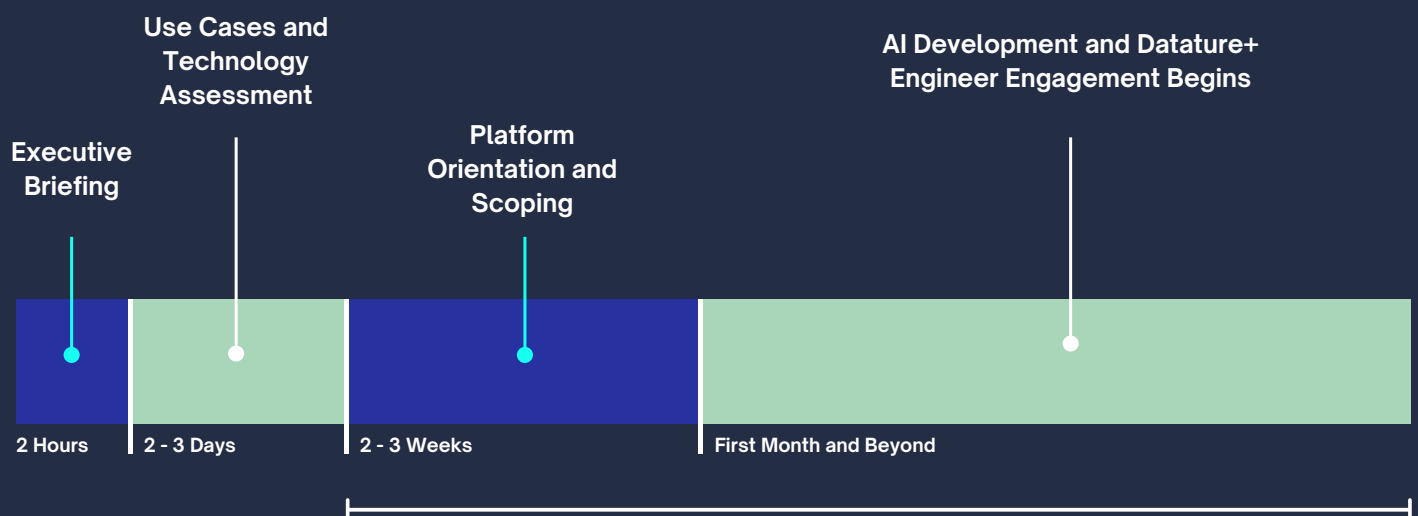
It comes with a myriad of toggles such as confidence, Intersection-over-Union, etc for one to check areas where the model might be lacking.

PLANS OVERVIEW

Updated as of 10 March 2022

	Developer	Professional	Professional+
Images	30,000	200,000	200,000
GPU Minutes	2,000	15,000	15,000
Users	3	5	10
Intellibrush Tokens	-	-	100,000

Datature Process Timeline



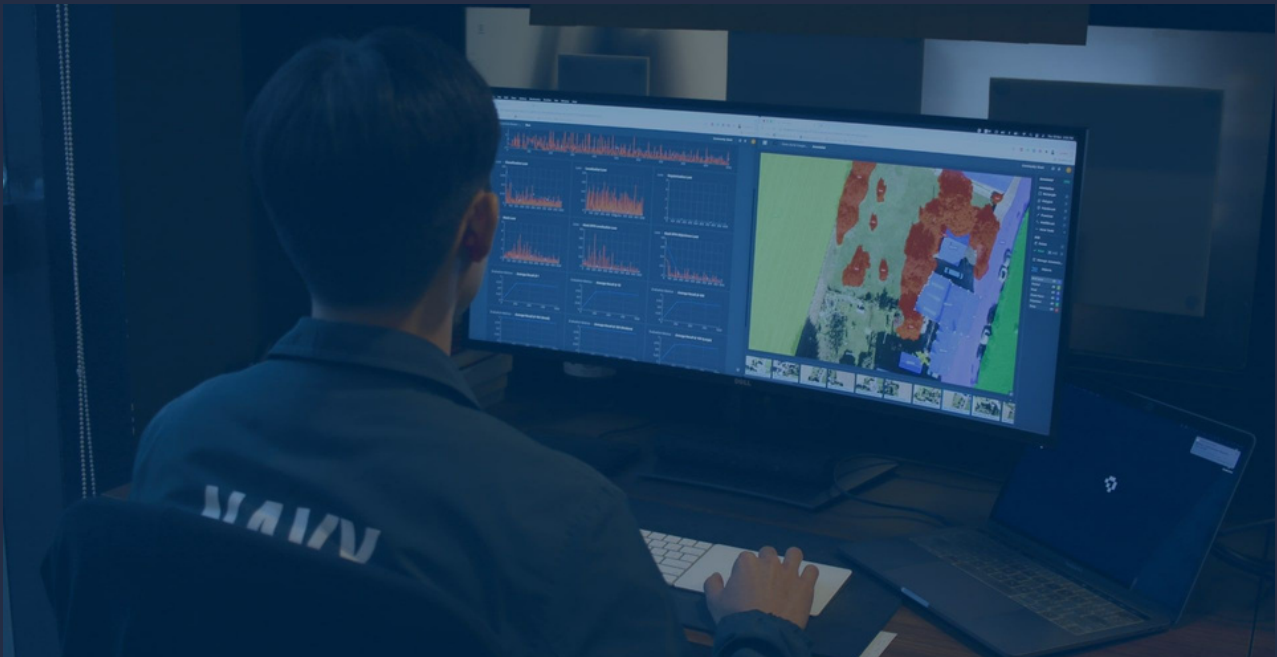
Clients understands how to use Datature to build more projects internally. Saving costs on redundant consulting - **Datature equip companies with the capabilities to build Enterprise AI internally.**

QUESTIONS? CONTACT US.

PRESENTED BY



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Co-founder | Datature



DATATURE

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[Datature Community Slack](#)